(a) Climate justice should be ingrained in detail in the rules of all the new climate compacts/ agreements.
(b) Environmental resources are unevenly distributed and exploited across the globe.
(c) There is an impending issue of dealing with a huge number of climate change victims/ climate refuges.
(d) Climate change in all its connotations is mostly due to developed countries and therefore their share of burden should be more.

Ans. (a) : Statement that best reflects the most logical, rational and crucial message conveyed by the passage is-
"Climate justice should be ingrained in detail in the rules of all the new climate compacts/agreements".

44. A principal P becomes Q in 1 year when compounded half-yearly with R% annual rate of interest. If the same principal P becomes Q in 1 year when compounded annually with S% annual rate of interest, then which one of the following is correct?
(a) R = S (b) R > S
(c) R < S (d) R ≤ S

Ans. (c) : According to the given,
P becomes Q in 1 year when compounded half yearly with R% annual rate of interest
Also, P becomes Q in the same time period of 1 year with S% annual rate of interest compounded annually
Whenever, Principle and amount are same for the same time period then, the rate of interest compound half yearly is always less than rate of interest compounded annually
Hence, R < S

45. How many natural numbers are there which give a remainder of 31 when 1186 is divided by these natural numbers?
(a) 6 (b) 7
(c) 8 (d) 9

Ans. (d) : According to the given, remainder should always be 31.
Therefore, 1186–31 = 1155
1155 will be exactly divisible by these numbers.

5	1155
11	231
7	21
	3

1155 = 5 × 11 × 7 × 3
Since, remainder should always be 31.
Therefore all the required numbers should be greater than 31.
(i) 3 × 11 = 33 (ii) 5 × 7 = 35
(iii) 5 × 11 = 55 (iv) 7 × 11 = 77
(v) 15 × 7 = 105 (vi) 15 × 11 = 165
(vii) 15 × 7 × 11 = 1155 (viii) 35 × 11 = 385
(ix) 77 × 3 = 231
So, total number of natural numbers is 9.

46. Let pp, qq and rr b 2-digit numbers where p < q < r. If pp + qq + rr = tt0, where tt0 is a 3-digit number ending with zero, consider the following statements :
1. The number of possible values of p is 5.
2. The number of possible values of q is 6.
Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : According to the given,
p < q < r
and pp + qq + rr = tto
Since unit digit of the given sum is zero.
Therefore
Case (i) Let p = 1

	p	q	r	
(a)	1	2	7	→ unit digit = 0
(b)	1	3	6	→ unit digit = 0
(c)	1	4	5	→ unit digit = 0

Case (ii) Let p = 2
Respective values of p, q and r is 2, 3 and 5.
2, 3, 5 → unit digit = 0
Case (iii) Let p = 3
Respective values of p, q and r is 3, 8 and 9.
3, 8, 9 → unit digit = 0
Case (iv) Let p = 4
Respective values of p, q and r is 4, 7 and 9.
4, 7, 9 → unit digit = 0
Case (v) Let p = 5
Respective values of p, q and r is 5,6,9
5,6,9 → unit digit = 0
None of the other values of p will fulfill the condition.
So, number of possible values of p is 5 (1,2,3,4,5) and number of possible values of q is 6 (2, 3, 4, 6, 7, 8).
Therefore both the statements 1 and 2 are correct.

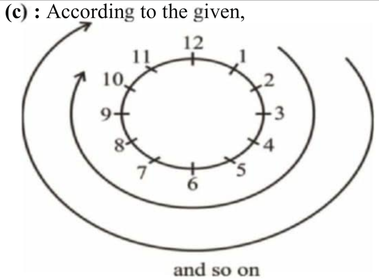
47. What is the sum of all 4-digit numbers less than 2000 formed by the digits 1, 2, 3 and 4, where none of the digits is repeated?
(a) 7998 (b) 8028 (c) 8878 (d) 9238

Ans. (a) : According to the given,
All four digit numbers which are less than 2000 and formed by digits 1, 2, 3, 4 are -

	1 2 3 4
	1 2 4 3
	1 3 2 4
	1 3 4 2
	1 4 2 3
+	1 4 3 2
	7 9 9 8

48. What is the number of selections of 10 consecutive things out of 12 things in a circle taken in the clockwise direction?
(a) 3 (b) 11 (c) 12 (d) 66

Ans. (c) : According to the given,



So, number of selections of 10 consecutive things from 12 things in clockwise direction =
(i) 1 to 10 (ii) 2 to 11 (iii) 3 to 12
(iv) 4 to 1 (v) 5 to 2 (vi) 6 to 3
(vii) 7 to 4 (viii) 8 to 5 (ix) 9 to 6
(x) 10 to 7 (xi) 11 to 8 (xii) 12 to 9
Hence, there are 12 possible number of selections.

49. If today is Sunday, then which day is it exactly on 10^{10}th day?
(a) Wednesday (b) Thursday
(c) Friday (d) Saturday

Ans. (b) : According to the given,
By dividing 10^{10} by 7 we will get the number of odd days.

$$\frac{10^{10}}{7}=\frac{\left(10^2\right)^5}{7}=\frac{(100)^5}{7}$$

$$\Rightarrow \text{Remainder} = \frac{2^5}{7}=\frac{32}{7} \Rightarrow \text{Remainder} = 4$$

Therefore, Remainder = odd days = 4
So, If today is Sunday, then 10^{10}th day will be Thursday

50. There are three traffic signals. Each signal changes colour from green to red and then form red to green. The first signal takes 25 seconds, the second signal takes 39 seconds and the third signal takes 60 seconds to change the colour from green to red. The durations for green and red colours are same. At 2 : 00 p.m, they together turn green. At what time will they change to green next, simultaneously?
(a) 4:00 p.m. (b) 4:10 p.m.
(c) 4:20 p.m. (d) 4:30 p.m.

Ans. (b) : According to the given,
1st signal takes time to change colour = 25 sec
2st signal takes time to change colour = 39 sec
3st signal takes time to change colour = 60 sec
L.C.M. of 25, 39 and 60 =

5	25, 39, 60
5	5, 39, 12
2	1, 39, 12
2	1, 39, 6
3	1, 39, 3
13	1, 13, 1
	1, 1, 1

LCM = 5 × 5 × 2 × 2 × 3 × 13 = 3900
So, all the three will become red after 3900 sec.
So, they will become green after another 3900 sec
Total time from green to green = 7800 sec

$$= \frac{7800}{60} = 130 \text{ min} = 2 \text{ hr } 10 \text{ min}$$

Therefore, again they will together become green at,
2 pm + 2 hr 10 min
= 4:10 PM

Directions for the followings 3 (three) items:
Read the following three passages and answer the items that follow the passages. Your answer to these items should be based on the passages only.

Passage–1

Sourcing food from non-agricultural lands (uncultivated system such as forests, wetlands, pastures, etc) in addition to agricultural lands enables a systemic approach to food consumption. It allows rural and tribal communities to sustain themselves for the whole year and steer clear of natural disasters and season-induced shortfalls of agricultural food. Since the productivity of trees is often more resilient to adverse weather conditions than annual crops, forest foods often provide a safety net during periods of food shortages caused by crop failure; forest foods also make important contributions during seasonal crop production gaps.

51. Which one of the following statements best reflects the most logical and rational message conveyed by the author of the passage?
(a) Food yielding trees should replace other trees in rural and tribal areas and community owned lands.
(b) Food security cannot be ensured in India with the present practice of conventional agriculture.
(c) Wastelands and degraded areas in India should be converted into agro forestry systems to help the poor.
(d) Agroecosystems should be developed in addition to or along with conventional agriculture.

Ans. (d) : Statement that best reflects the most logical and rational message conveyed by the author of the passage is-
Agroecosystems should be developed in addition to or along with conventional agriculture.

Passage–2

While awareness on use/misuse and abuse of antibiotics is common knowledge, as is the impact of dosing poultry with antibiotics, the environmental impact of antibiotics-manufacturing companies not treating their waste has scarcely been discussed at any length or seriousness thus far. Pollution from antibiotics factories is fueling the rise of drug-resistant infections. The occurrence of drug-resistant bacteria surrounding the pharma manufacturing plants is well known.

52. Which one of the following statements best reflects the most logical and practical message conveyed by the passage?
(a) It is necessary to put proper effluent treatment protocols in place.
(b) It is necessary to promote environmental awareness among people.
(c) Spread of drug-resistant bacteria cannot be done away with, as it is inherent in modern medical care.
(d) Pharma-manufacturing companies should be set up in remote rural areas, away from crowded towns and cities.

Ans. (a) : Statement that best reflects the most logical and practical message conveyed by the passage is- "It is necessary to put proper effluent treatment protocols in place."

Passage–3

Benefits of good quality school education accrue only when students complete and leave school after having acquired the gateway skills. Like one learns to